React Introduction (Detailed Guide for PDF)

What is React?

React is a JavaScript library used for building user interfaces, especially single-page applications (SPA).

History of React

- Developed by Facebook (Meta)
- Released in 2013
- Maintained by Meta and open-source community

Why Use React?

- Fast performance
- Reusable components
- Easy to maintain
- Strong community support

Features of React

- Component-based architecture
- Virtual DOM for fast rendering
- One-way data binding
- JSX (JavaScript XML)

JSX (JavaScript XML)

```
const element = <h1>Hello React</h1>;
```

Components

Functional Component

```
function App() {
return <h1>Hello</h1>;
}
```

Props

Used to pass data to components.

```
function Welcome(props) {
return <h1>Hello {props.name}</h1>;
}
```

State

```
import { useState } from "react";

function Counter() {
const [count, setCount] = useState(0);
}
```

Event Handling

```
<button onClick={() => alert("Clicked")}>Click</button>
```

Conditional Rendering

```
{isLoggedIn ? <h1>Welcome</h1> : <h1>Please Login</h1>}
```

Lists and Keys

```
const list = [1,2,3];
list.map(item => <li key={item}>{item}</li>);
```

React Hooks

useState

useEffect

```
useEffect(() => {
console.log("Component Loaded");
}, []);
```

Forms in React

```
<input type="text" onChange={(e) => console.log(e.target.value)} />
```

Routing (React Router)

Used for navigation.

Example Project

```
function App() {
return (
<div>
<h1>Hello React</h1>
</div>
);
}
```

Advantages of React

- Fast and efficient
- Reusable components
- Large ecosystem

Disadvantages of React

- Requires learning JSX
- Frequent updates

Best Practices

- Use functional components
- Keep components small
- Use hooks properly